

FIORI X AI

AI-assisted migration of legacy free-style Fiori apps to best practices

WHO AM I?

- Freelancer
- Python + Javascript dev
- 20y SAP BC
- Heavy Claude Code user:
 - SAP BC automation
 - SAP CAP in BTP
 - SAP HCM + Fiori on-premise

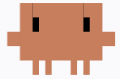
AGENDA

- Introduction to Claude Code
- SAP HCM Portal use case
- SAPUI5 best practices

CLAUDE CODE

- Just a tool
 - Helps you accelerate your work
 - Vibecode

CLAUDE CODE



Claude Code v2.1.277

Fable 5.1 with medium effort · Claude Max

~\Documents\git\hcmdemo

> show me a diagram with the flow of between the views of this fiori app

- I'll read the routing config and the navigation calls in the controllers, then draw the flow.

Ran 2 shell commands

- One route, one view. The flow is between the Main view and its fragments. I'll look at which handlers open which fragment and what leads where.

Searched for 1 pattern, ran 1 shell command

- I need the section visibility bindings and what each dialog handler does after confirm. One call.

Ran 2 shell commands

- I still need the section visibility bindings and which handlers set the flags that switch sections. One call.

Ran 1 shell command

- One route, one view. The app has no view-to-view navigation. The flow is between the four sections of the Main ObjectPage and the 15 fragments it opens.

```
App.view (root, NavContainer "app")
└─ route "" → Main.view (ObjectPageLayout)
```

```
    Section 1 Administration (always visible)
      ComboBox change → IsAdminSelected = true
```

```
    Section 2 Déclaration ← visible = IsAdminSelected
      [Ouvrir] → OpenRequest (TableSelectDialog)
                  confirm → SelectedRequest, IsActionSelected = true
      [Créer] → POST RequestSet → IsActionSelected = true
```

CLAUDE CODE

- Just a tool
 - Helps you accelerate your work
 - ~~Vibecode~~
 - Documentation, retro-engineering
 - Troubleshooting
 - Automation
 - Performance optimisation
 - Testing

EXAMPLES OUTSIDE OF “WRITING APPS”

- Dig through spaghetti ABAP code retro-engineering
- Analyse sapui5 config relevance retro-engineering
- Babysit deployments (Cloud Foundry, Nomad, etc..) automation
- Interpret Cloud Foundry audit logs troubleshooting
- Extensions to the ABAP MCP enabling AI to easily : automation
 - Create SM30 maintenance views
 - Create SEGW projects (full support)
 - Create DDIC tables (NW 7.50)
 - Create DDIC domains (NW 7.50)
 - Configure FLP groups and tiles
- Write SAPUI5 Playwright E2E tests testing
- Optimise Fiori OData \$batch grouping performance

UI5-LOCAL.YAML

```
! ui5-local.yaml
1 # yaml-language-server: $schema=https://sap.github.io/ui5-tooling/schema/ui5.yaml.json
2 # The daily driver: SAPUI5 from the local runtime build, backend through the proxy.
3 # Same as ui5.yaml apart from the framework source. Edit both together.
4
5 specVersion: "4.0"
6 metadata:
7   name: lu.bouge.demo
8   type: application
9 server:
10   customMiddleware:
11     # Transpile webapp/**/*.ts on the fly, so JS controllers require a shared module
12     # under its extensionless path exactly as they require a .js one. Every serve config
13     # needs it: a config missing it answers 404 and the app dies at load.
14     - name: ui5-tooling-transpile-middleware
15       afterMiddleware: compression
16     # Framework from the local runtime build (bundles included – the npm framework
17     # packages ship sources only). Shared by every project that mounts it. Mounted at /
18     # so /resources and /test-resources resolve inside it; unmatched paths fall through.
19     - name: ui5-middleware-servestatic
20       afterMiddleware: compression
21       mountPath: /
22       configuration:
23         rootPath: "C:/ui5-runtime/1.108.55"
24     - name: fiori-tools-proxy
25       afterMiddleware: compression
26       configuration:
27         ignoreCertError: true
28         backend:
29           - path: /sap
30             url: https://<FLP SERVER>
31             client: '140'
32     - name: fiori-tools-appreload
33       afterMiddleware: compression
34       configuration:
35         port: 35729
36         path: webapp
37         delay: 300
38     - name: fiori-tools-preview
39       afterMiddleware: fiori-tools-appreload
40       configuration:
41         flp:
42           path: /test/flp.html
43           theme: "<THEME>@/sap/public/bc/themes/~client-140"
44         intent:
45           object: Remuneration
46           action: requestClothingCompensation
47
```

HOW GOOD ARE AI AGENTS AT

Writing <i>correct</i> code	++	
Code architecture	+	
Writing tests	+	
Migrating code	+++	
Analysing code	+++	

CURRENT SITUATION

- SAP HCM ESS Portal based on Fiori
- 80+ apps developed over the last 10 years
- Many apps have not be touched in the last 5+ years
 - (except for small fixes due to upgrades)
- Mostly freestyle apps
- Development on local VSCode (WebIDE previously)

CURRENT SITUATION - PROBLEMS

- Components/libraries copy/pasted across apps
- Each developer with their own preferences, their own laptop setup
- Limited testing
- No overview of JS dependencies

“BEST PRACTICES” BASELINE

- Initialise CLAUDE.md + shared KB } Phase 1
- Remove WebIDE legacy config
- Align i18n config
- Align ui5-*.yaml config
- flpSandbox config from live FLP config
- Enable Typescript
- Configure linter + formatter : UI5 linter, Prettier, oxlint } Phase 2
- Format code } Phase 3
- Wire in the shared library (ODataHelper.ts + a few components) } Phase 4

(1) GENERATE CLAUDE.MD

Skill /sap-clause-prompt-simplify

- Based off official claude-api/prompt-audit
- Integrates my own SAP KB distributed other several .md files
- Identify per projects which KB files apply or not
- At the end, calls “claude -p” with sonnet to verify the agent knows how to fetch info from the KB

• Done: **CLAUDE.md** created, all 3 verification tests pass

1. UI5 trigger: pass – /sap-ui5 invoked, ui5-freestyle.md "Binding" read.
2. Loaded rule: pass – rule 1 quoted exactly from the project CLAUDE.md, new pointer block quoted.
3. ABAP trigger: pass – /sap-abap invoked, segw.md and nw750-quirks.md sections read.

File	Change
CLAUDE.md	New: project paragraph, skills pointer block, status marker 2026-09-17 applied applied, always-on UI5 rules (4, 6, 8 dropped), bootstrapping paragraph.

(1) GENERATE CLAUDE.MD

```
1 # hcmdemo – CLAUDE.md
```

```
2
3 Demo UI5 1.108 Freestyle app (plain JavaScript), namespace `lu.bouge.demo`. Started from a copy of [REDACTED]; the
4 `manifest.json` / `ui5-deploy.yaml` are inherited from it and not yet the demo's own.
```

5 ## Shared knowledge base

```
6
7 The shared SAP rules are skills, not imports. Each loads a table of contents on invocation and reads its reference files by section:
```

- ```
8
9 - **ABAP task** (BL class, CDS, DPC_EXT, SEGW, ADT tools, transports, HCM tables): run `/sap-abap` first.
10 - **UI5 task** (anything under `webapp/`): run `/sap-ui5` first.
11 - Run `/sap-learn` at the end of a session to route new rules to the shared files and project facts to this file.
```

```
12
13 claude-prompt-simplify: 2026-09-17 applied applied
```

```
14 ui5-baseline: 2026-09-18 integrate-shared-libs integrated
```

```
15 ui5-lint: 2026-09-18 ui5lint 226 errors, 74 warnings, oxlint 47 errors, 9 warnings
```

```
16 ui5-format: 2026-09-18 clean
```

## 17 ### Rules that apply to every UI5 edit

```
18
19
20 Copied from `~/ .claude/skills/sap-ui5/SKILL.md`. Rules 4 and 8 dropped (plain JavaScript, no TypeScript). Rule 6 dropped (no embed component)
```

## (2) BASELINE MIGRATION – CONFIG ONLY

```
hcmdemo
├── .che
├── dist
├── node_modules
├── webapp
│ ├── controller
│ ├── css
│ ├── i18n
│ │ └── i18n.properties
│ ├── lib
│ ├── localService
│ ├── model
│ └── test
│ ├── Component-changes.json
│ ├── fakeLRep.json
│ ├── flpSandbox.html
│ ├── flpSandboxMockServer.html
│ ├── locate-reuse-libs.js
│ ├── testFLPService.html
│ └── testFLPServiceMockServer.html
│ ├── view
│ ├── Component.js
│ ├── index.html
│ ├── manifest.json
│ └── serviceBinding.js
├── .eslintrc
├── .eslintrc.ext
├── .gitignore
├── CLAUDE.md
├── neo-app.json
├── package-lock.json
├── package.json
├── ui5-deploy.yaml
├── ui5-local.yaml
└── ui5.yaml
```

=>

```
hcmdemo
├── dist
├── node_modules
├── webapp
│ ├── appconfig
│ │ └── fioriSandboxConfig.json
│ ├── controller
│ ├── css
│ ├── i18n
│ │ └── i18n_fr.properties
│ ├── lib\shared
│ ├── localService
│ ├── model
│ ├── view
│ ├── Component.js
│ ├── index.html
│ ├── manifest.json
│ ├── .gitignore
│ ├── .gitmodules
│ ├── .oxlintrc.json
│ ├── .prettierrc.json
│ ├── CLAUDE.md
│ ├── package-lock.json
│ ├── package.json
│ ├── tsconfig.json
│ ├── ui5-deploy.yaml
│ ├── ui5-dist.yaml
│ ├── ui5-local.yaml
│ ├── ui5-mock.yaml
│ ├── ui5.yaml
│ └── ui5lint.config.mjs
```

Skill /sap-ui5-baseline

Includes basic checks to make sure the environment is fully ready for the developer (start fiori dev etc...)

No code change (only config)

### Counts (nothing fixed)

| Tool     | Result                  |
|----------|-------------------------|
| ui5lint  | 226 errors, 74 warnings |
| oxlint   | 63 errors, 11 warnings  |
| prettier | 29 files unformatted    |

## (2) BASELINE MIGRATION – CONFIG ONLY

---

- Planning w/ Fable, execution w/ Opus 5 subagent

```
• ui5-baseline-migrator(Run UI5 baseline migration) Opus 5 (1M context)
 └ Backgrounded agent (↓ to manage • ctrl+o to expand)

• Migrator is running. I will verify its tree (step 7) and report when it finishes.
```

- 10 min ~6 eur API tokens  
(Claude Max 180e = ~6000e of API tokens)
- Cost can be optimised
- Fable is good at handling deviations from the plan
- Opus is good at changing only what is asked

### (3) FORMATTING

---

- `npm run format`
- Formatting updates go in a separate commit

## (4) BASELINE MIGRATION – CODE

### • Phase 2: integrated, committed as 222c648, verified

Every 7b check passes on the tree: `ts-check` clean, build clean, `dist/lib/shared/src/` holds only `.js` + `-dbg.js` + maps plus `wf/` assets and the vendored `xlsx` trio (no `-dbg.js`, no `.map` there), the served module paths answer 200, the exit-criterion `grep` and the external-script `grep` both return nothing. Working tree clean.

#### Check in the app (1)

1. Drive the 13 paths that now go through `this.odata()`. App start (`GetDefaultAdmin`), create a declaration, open one (`GetTotals` and `GetMessCount` on each list refresh), import a list, add agents from the search dialog, upload Excel file 1 and file 2 (the `xlsx` library now loads on demand from the submodule), delete selected agents, edit a header field, edit an agent row, submit, return, delete the declaration.

#### Decide (2)

2. `xlsx 0.10.0` from `cdnjs` is now `SheetJS CE 0.20.3` vendored in `ui5-shared`. The migrator checked that `XLSX.read` with `type: "binary"` and `sheet_to_row_object_array` still exist. The two upload paths are the place a behaviour difference would show. The `jszip.js` script was dropped because the full build carries its own zip code.
3. `_onSubmitRequestSAVE` and `_onReturnRequestSAVE` look dead. They were migrated but no view binds them. Delete them or keep them; a separate unit.

#### Changed files

| File                                                 | Change                                                                                                                                                                                                     |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>webapp/controller/BaseController.js</code>     | Adds <code>odata()</code> and <code>notifyODataError()</code> .                                                                                                                                            |
| <code>webapp/controller/Main.controller.js</code>    | 16 raw <code>OData</code> calls onto <code>this.odata()</code> ; error blocks onto <code>notifyODataError</code> ; <code>XLSX</code> global replaced by an on-demand load in <code>_readExcelRows</code> . |
| <code>webapp/controller/ErrorHandler.js</code>       | Deleted.                                                                                                                                                                                                   |
| <code>webapp/Component.js</code>                     | Drops <code>ErrorHandler</code> wiring and the two CDN script tags; registers the <code>xlsx</code> loader path.                                                                                           |
| <code>webapp/i18n/i18n_fr.properties</code>          | Removes <code>errorTitle</code> , <code>errorText</code> .                                                                                                                                                 |
| <code>ui5.yaml</code> , <code>ui5-deploy.yaml</code> | Vendor opt-in: <code>componentPreload.excludes</code> and <code>minification.excludes</code> .                                                                                                             |
| <code>webapp/lib/shared</code>                       | Pointer <code>3c4a002</code> → <code>ee745c7</code> , which carries <code>src/vendor/xlsx/</code> .                                                                                                        |
| <code>CLAUDE.md</code>                               | New markers; rule 3 no longer says raw calls remain; one paragraph on the <code>xlsx</code> opt-in.                                                                                                        |

Markers now read `ui5-baseline: 2026-09-18 integrate-shared-libs integrated`, `ui5-lint: ui5lint 226 errors, 74 warnings, oxlint 47 errors, 9 warnings`, `ui5-format: clean`. No lint finding was fixed; the `oxlint` drop comes from the deleted `ErrorHandler` and removed unused callback parameters.

#### Corrections

Two behaviour notes from the migrator. Deferred deletes in `delAgentsGroup` get an empty `.catch` so the `submitChanges` promise stays the single error path, as before. The two `FileReader.onerror` handlers used to throw silently on `JSON.parse` of a DOM event; they now log and show "Lecture du fichier impossible." That is the one non-`OData` behaviour change. The old format marker of 29 files was stale: `commit 0dfab75` had already formatted the repo, so `format:check` is clean.

## (4) BASELINE MIGRATION – CODE

---

- Code changes so it requires testing
- By default agent mimick surrounding code (including legacy practises)

```
oModel.update(changedItemPath, changedItemData, {
 success: function (oSuccess) {
 MessageToast.show("Modification enregistrée");
 },
 error: function (oError) {
 MessageToast.show("Problème rencontré");

 var oFuncError = JSON.parse(oError.responseText);
 MessageBox.error(oFuncError.error.message.value);
 },
});
```



```
this.oData()
 .update(changedItemPath, changedItemData)
 .then(function () {
 MessageToast.show("Modification enregistrée");
 })
 .catch(function (oError) {
 MessageToast.show("Problème rencontré");

 that.notifyODataError(oError, "update request header");
 });
```



```
try {
 await this.oData().update(changedItemPath, changedItemData);
 MessageToast.show("Modification enregistrée");
} catch (oError) {
 MessageToast.show("Problème rencontré");
 this.notifyODataError(oError, "update request header");
}
```

# NEXT STEPS

---

- Fix legacy sapui5 syntax (cf. ui5 linter)
- Modernise syntax / full Typescript migration
- Write E2E tests
- Security review (frontend + backend code)

# THANK YOU

---

sami@bouge.lu

# BASELINE.MD

```
1 # UI5 project baseline ☞ the standard config layout, and how to migrate onto it
2
3 Decisions and ruled-out alternatives: `ui5-project-baseline.decisions.md`, next to this file. Do not load it here; read it only to change this procedure.
4
5 The file and config layout every Freestyle app on this stack should carry. It is one procedure,
6 applied once per app. New apps start here; existing apps are migrated with the steps below.
7
8 The *rules* behind each choice already live in `ui5-freestyle.md` ☞ *Local dev server*,
9 *TypeScript on UI5 1.108*, *FLP intents & startup parameters*, and in `ui5-shared.md`. **Do not
10 re-derive them and do not restate them here** ☞ this file is the procedure and the exact file
11 bodies, nothing else.
12
13 **Running it** ☞ one prompt in the target project:
14
15 ```
16 Bring this app onto the UI5 project baseline ☞ follow ~/.claude/skills/sap-ui5/ui5-project-baseline.md.
17 ```
18
19 > ## Target state ...
20
49
50 > ## Phases ...
51
63
64 > ## Parameters ☞ read these out of the project before starting ...
65
76
77 > ## 1. Toolchain ...
78
99
100 > ## 2. The five yaml files ...
101
305
306 > ## 3. FLP intent and tile ...
307
397
398 > ## 4. `package.json` scripts ...
399
420
421 > ## 5. i18n ...
422
469
470 ## 6. The shared submodule
471
472 > ### 6a. Mount (phase 1) ...
473
505
506 > ### 6b. Integrate (phase 2) ...
507
517
518 ## 7. Verify, in this order
519
520 > ### 7a. Phase 1 ☞ every step below ...
521
573
574 > ### 7b. Phase 2 ...
575
584
585 > ## 8. Linting ...
586
675
676 > ## 9. Formatting ...
677
729
730 > ## Known traps ...
731
753 |
```